Monte Carlo Coulomb Collisions for Guiding-Centre Particles in Boozer Coordinates: Implementation Notes

firm3d — `ep_thermal_collisions`

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1 Overview

These notes describe the algorithm used in `trace_particles_boozer_with_collisions` to include stochastic Coulomb collisions for an energetic particle (EP) tracing through one or more Maxwellian background species.

The approach couples the deterministic guiding-centre (GC) orbit equations in Boozer coordinates with a Milstein stochastic term for the velocity-space diffusion. The integrated state is the ordinary GC state $(s, \theta, \zeta, v_\parallel)$ at fixed magnetic moment μ; the collision operator acts on the velocity-space pair (v, ξ), where $v = |\boldsymbol{v}|$ is the total speed and $\xi = v_\parallel / v$ the pitch, as a kick applied between orbit steps (Section 6.3). Because μ is constant across an orbit step it is a parameter of the orbit equations rather than a state variable, which is what allows the vacuum, $K = 0$ and full GC equations all to be driven by the same collision operator.

Primary references:

- Hirvijoki et al. [1] — Monte Carlo implementation of the guiding-centre Fokker–Planck equation in (v, ξ) coordinates; Eqs. (32)–(33) of that paper define the SDE system used here.
- Hirvijoki et al. [2] — ASCOT4/5 code description; source file `mccc_coefs.h` defines the Coulomb logarithm convention adopted here.
- Boozer & Kuo-Petravic [4] — pitch-angle representation in Boozer coordinates.

2 Guiding-Centre Equations in Boozer Coordinates

In Boozer coordinates (s, θ, ζ), where s is the normalised toroidal flux, θ the poloidal angle, and ζ the toroidal angle, the vacuum GC equations are

$$\dot{s} = -\frac{m\, v^2(1+\xi^2)}{2\, q\, \psi_0} \frac{\partial B}{\partial \theta} \frac{1}{B}, \tag{1}$$

$$\dot{\theta} = \frac{m\, v^2(1+\xi^2)}{2\, q\, \psi_0} \frac{\partial B}{\partial s} \frac{1}{B} + \frac{\iota\, \xi\, v\, B}{G}, \tag{2}$$

$$\dot{\zeta} = \frac{\xi\, v\, B}{G}, \tag{3}$$

$$\dot{\xi} = -\frac{(\iota\, \partial_\theta B + \partial_\zeta B)\, v\, (1-\xi^2)}{2G} \quad \text{[mirror force]}, \tag{4}$$

where ψ_0 is the reference poloidal flux, $G(s)$ is the covariant toroidal field component, and $\iota(s)$ is the rotational transform. The magnetic moment $\mu = v^2(1-\xi^2)/(2B)$ is held fixed through an orbit step and updated by the collision kick. The vacuum equations are shown here for concreteness; the $K = 0$ and full GC equations of `trace_particles_boozer` are supported identically and are selected from the field's `field_type`.

With collisions the equations for v and ξ acquire deterministic drift terms (Section 4) and stochastic noise terms (Section 5.2).

3 Background Species and Profiles

Each background species b is described by a `ThermalBackground` object specifying:

- $n_b(s)$ — number density [m^{-3}], callable of the flux label $s \in [0, 1]$;
- $T_b(s)$ — temperature [J], callable (use `T_keV` $\times$ `1e3` $\times e$);
- m_b — species mass [kg];
- q_b — species charge [C] (signed).

Internally the profiles are pre-evaluated on a uniform s-grid of 512 points and linearly interpolated in C++.

4 Collision Coefficients

Notation

The symbols used here follow the ASCOT5 source code (`mccc_coefs.h`, [3]); Table 1 shows the correspondence with Hirvijoki et al. [1], whose Eqs. (32)–(33) define the SDE structure.

This paper	ASCOT5 `mccc_coefs.h`	Hirvijoki et al. [1]
$D_\parallel$	`Dpara`	$D_k/2$
ν_D	`nu` $= 2D_\perp/v^2$	$2D_\perp/v^2$
Q	`Q`	(Einstein-relation drag in K)
K	`K` $= Q + D'_\parallel + 2D_\parallel/v$	K (drift in Eq. 32)

Table 1: Notation correspondence. The noise amplitude for v is $g_v = \sqrt{2D_\parallel}$ (this paper) $= \sqrt{D_k}$ (Hirvijoki 2013).

4.1 Debye Length

The total Debye screening length from all background species is [5, 6]

$$\frac{1}{\lambda_D^2} = \sum_b \frac{n_b\, q_b^2}{\varepsilon_0\, T_b}. \tag{5}$$

4.2 Coulomb Logarithm

The Coulomb logarithm is defined as the ratio of maximum to minimum impact parameters [5, 6, 7]:

$$\ln\Lambda_{ab} = \ln\left(\frac{\lambda_D}{b_{\min}}\right), \qquad b_{\min} = \max(b_{\mathrm{cl}},\, b_{\mathrm{qm}}). \tag{6}$$

The maximum impact parameter is the Debye length (Eq. (5)). The minimum impact parameter is the larger of the classical 90°-deflection distance and the quantum-mechanical de Broglie wavelength of the reduced mass:

$$b_{\mathrm{cl}} = \frac{|q_a\, q_b|}{4\pi\varepsilon_0\, m_r\, \bar{v}^2}, \tag{7}$$

$$b_{\mathrm{qm}} = \frac{\hbar}{2\, m_r\, \sqrt{\bar{v}^2}}, \tag{8}$$

where $m_r = m_a m_b/(m_a + m_b)$ is the reduced mass.

The effective velocity scale $\bar{v}^2$ is taken as

$$\bar{v}^2 = v^2 + v_{\mathrm{th},b}^2, \qquad v_{\mathrm{th},b} = \sqrt{2T_b/m_b}, \tag{9}$$

following the ASCOT5 source convention [3]. This form is a numerical interpolation device, not derived from first principles: Ichimaru [7] derives b_{cl} and b_{qm} only in the fast-particle limit $v \gg v_{\mathrm{th},b}$ (Section 4.3B of that reference), where $\bar{v} \approx v$. The $\bar{v}^2$ form extends the formula to the slow-particle regime without a discontinuous switch: fast EP against ions ($v \gg v_{\mathrm{th},b}$, $\bar{v} \approx v$) and slow EP against electrons ($v \ll v_{\mathrm{th},e}$, $\bar{v} \approx v_{\mathrm{th},e}$).

The quantum branch $b_{\mathrm{qm}} > b_{\mathrm{cl}}$ applies when

$$\frac{2\pi\varepsilon_0\, \hbar\, \sqrt{\bar{v}^2}}{|q_a\, q_b|} > 1, \tag{10}$$

i.e. when the de Broglie wavelength of the reduced mass particle exceeds the classical distance of closest approach. For 3.52 MeV α-particles scattering off 10 keV electrons one finds $b_{\mathrm{qm}} \approx 6.9\, b_{\mathrm{cl}}$, so the quantum branch is active throughout alpha slowing-down against electrons. If $\ln\Lambda \leq 0$ (Debye length below the minimum impact parameter, e.g. $T_b \to 0$ at finite density) the binary-collision model is undefined and an exception is raised; ASCOT5 reaches the equivalent outcome by aborting markers through its input-evaluation errors, and its input profiles keep the temperature finite wherever the density is nonzero. A warning is issued for $0 < \ln\Lambda < 2$, where the perturbative expansion underlying the Fokker–Planck operator begins to lose validity.

Both checks are applied to the profiles once, before tracing starts, rather than from inside the right-hand side: the latter is evaluated roughly seven times per accepted step per particle, so a per-evaluation warning would emit millions of duplicate lines, and a per-evaluation exception would fire on whichever MPI rank owns the offending particle, hanging the remaining ranks in the collective that gathers results. Since $\ln\Lambda$ increases monotonically with v through $\bar{v}^2$, evaluating at $v \to 0$ bounds it from below over all particle speeds, making the up-front check conservative but never optimistic.

4.3 Chandrasekhar G Function

The Chandrasekhar function [8] is

$$G(x) = \frac{\operatorname{erf}(x) - \dfrac{2x}{\sqrt{\pi}}\, e^{-x^2}}{2x^2}, \qquad x = v/v_{\mathrm{th},b}. \tag{11}$$

Its derivative satisfies the recurrence

$$G'(x) = \frac{2}{\sqrt{\pi}}\, e^{-x^2} - \frac{2\,G(x)}{x}. \tag{12}$$

Key limiting behaviours:

$$\begin{aligned} x \to 0: &\quad G \to 0, \quad G' \to 2/(3\sqrt{\pi}) \approx 0.376; \\ x \to \infty: &\quad G \to 1/(2x^2), \quad G' \to 0. \end{aligned}$$

4.4 Collision Frequency Prefactor

Define the Rosenbluth prefactor for species b [9, 10]:

$$\Gamma_b = \frac{n_b\, q_a^2\, q_b^2\, \ln \Lambda_{ab}}{4\pi\varepsilon_0^2\, m_a^2}. \tag{13}$$

4.5 Pitch-Angle Scattering Frequency

$$\nu_D^{(b)} = \frac{\Gamma_b\, [\operatorname{erf}(x) - G(x)]}{v^3} \;\equiv\; \frac{2D_\perp^{(b)}}{v^2}. \tag{14}$$

The second form uses the perpendicular diffusion coefficient $D_\perp^{(b)} = \Gamma_b[\operatorname{erf}(x) - G(x)]/(2v)$ and matches ASCOT5's `mccc_coefs_nu` and the $2D_\perp/v^2$ notation of Hirvijoki et al. [1, 2].

4.6 Parallel Diffusion Coefficient

$$D_\parallel^{(b)} = \frac{\Gamma_b\, G(x)}{v}, \qquad \frac{\partial D_\parallel^{(b)}}{\partial v} = \frac{\Gamma_b}{v}\left(\frac{G'(x)}{v_{\mathrm{th},b}} - \frac{G(x)}{v}\right). \tag{15}$$

In Hirvijoki et al. [1] the parallel diffusion coefficient is denoted D_k, with $D_\parallel = D_k/2$; the noise amplitude for the speed SDE is $\sqrt{D_k} = \sqrt{2D_\parallel}$ in both conventions. ASCOT5 names this coefficient `Dpara`. The v-derivative is required by the Milstein correction term (Section 5.2). Note that it differentiates Γ_b as though $\ln \Lambda_{ab}$ were constant, even though the $\ln \Lambda_{ab}$ actually evaluated depends on v through $\bar{v}^2$; this follows ASCOT5 and has a small but systematic consequence for the equilibrium, quantified in Section 10.

4.7 Deterministic Drag in v

The drag coefficient (ASCOT5: `mccc_coefs_Q` [3]) is fixed by the Einstein relation:

$$Q^{(b)} = -\frac{m_a\, v}{T_b}\, D_\parallel^{(b)} = -\Gamma_b\, \frac{m_a\, G(x)}{T_b}. \tag{16}$$

The total deterministic drift in speed is

$$K^{(b)} = Q^{(b)} + \frac{\partial D_\parallel^{(b)}}{\partial v} + \frac{2\,D_\parallel^{(b)}}{v} \tag{17}$$

(ASCOT5's `mccc_coefs_K` [3]; the drift coefficient K of Eq. (32) in Hirvijoki et al. [1]).

Itô transformation leading to Eq. (17)

Equation (17) can be obtained two independent ways, which cross-check each other.

(i) Itô's formula on $v = |\boldsymbol{v}|$. In three-dimensional velocity space the collisional motion of the test particle is

$$\mathrm{d}\boldsymbol{v} = F_{\mathrm{dyn}}\,\hat{\boldsymbol{v}}\,\mathrm{d}t + \boldsymbol{\sigma}\cdot\mathrm{d}\boldsymbol{W}, \qquad \boldsymbol{\sigma}\boldsymbol{\sigma}^\top = 2D_\parallel\,\hat{\boldsymbol{v}}\hat{\boldsymbol{v}} + 2D_\perp\,(\mathsf{I} - \hat{\boldsymbol{v}}\hat{\boldsymbol{v}}), \tag{18}$$

where F_{dyn} is the dynamical friction [8, 10, 11]. The speed $v = |\boldsymbol{v}|$ is a nonlinear function of $\boldsymbol{v}$, so Itô's formula applies with $\nabla_{\boldsymbol{v}}\,v = \hat{\boldsymbol{v}}$ and Hessian $\nabla_{\boldsymbol{v}}\nabla_{\boldsymbol{v}}\,v = (\mathsf{I} - \hat{\boldsymbol{v}}\hat{\boldsymbol{v}})/v$:

$$\mathrm{d}v = \Big(F_{\mathrm{dyn}} + \frac{2D_\perp}{v}\Big)\mathrm{d}t + \sqrt{2D_\parallel}\,\mathrm{d}W_v. \tag{19}$$

The $2D_\perp/v$ drift is the Bessel-process effect: kicks transverse to $\boldsymbol{v}$ do not change v at first order, but they lengthen the vector at second order, $\Delta v = |\boldsymbol{v} + \Delta\boldsymbol{v}_\perp| - v \approx |\Delta\boldsymbol{v}_\perp|^2/2v > 0$.

(ii) Fokker–Planck matching (detailed balance). The Itô SDE $\mathrm{d}v = K\,\mathrm{d}t + \sqrt{2D_\parallel}\,\mathrm{d}W_v$ evolves the speed density $P(v) \propto v^2 f(v)$ according to $\partial_t P = -\partial_v(KP) + \partial_v^2(D_\parallel P)$. Zero flux on the equilibrium $P_{\mathrm{eq}} \propto v^2\,e^{-m_a v^2/2T_b}$ requires $KP_{\mathrm{eq}} = \partial_v(D_\parallel P_{\mathrm{eq}})$, i.e.

$$K = -\frac{m_a v}{T_b}\,D_\parallel + \frac{\partial D_\parallel}{\partial v} + \frac{2\,D_\parallel}{v}, \tag{20}$$

which is Eq. (17) with Q as in Eq. (16). Here $\partial_v D_\parallel$ is the multiplicative-noise (Itô) drift that appears when the diffusion term is commuted into advective form, and $2D_\parallel/v$ comes from the Jacobian v^2 of spherical velocity-space coordinates.

Consistency of (i) and (ii). For Maxwellian field particles the dynamical friction obeys the identity

$$F_{\mathrm{dyn}}^{(b)} = -\Big(1 + \frac{m_a}{m_b}\Big)\frac{2x^2\,G(x)\,\Gamma_b}{v^2} = Q^{(b)} + \frac{\partial D_\parallel^{(b)}}{\partial v} + \frac{2\,(D_\parallel^{(b)} - D_\perp^{(b)})}{v}, \tag{21}$$

so that $F_{\mathrm{dyn}} + 2D_\perp/v = Q + \partial_v D_\parallel + 2D_\parallel/v = K$ exactly. (F_{dyn} is ASCOT5's `mccc_coefs_F`, which is not used in the guiding-centre scheme.) In the fast-particle limit $x \gg 1$ this identity encodes the classic cancellation

$$F_{\mathrm{dyn}} + \frac{2D_\perp}{v} \;\longrightarrow\; -\Big(1 + \frac{m_a}{m_b}\Big)\frac{\Gamma_b}{v^2} + \frac{\Gamma_b}{v^2} = -\frac{m_a}{m_b}\,\frac{\Gamma_b}{v^2} \;:$$

most of the dynamical friction is compensated by transverse diffusion, and the net slowing-down rate is set by the mass ratio. Note in particular that Q is *not* the dynamical friction — it is the Einstein-relation drag $-(m_a v/T_b)D_\parallel$.

4.8 Summation over Species

All coefficients are additive over background species:

$$\nu_D = \sum_b \nu_D^{(b)}, \quad D_\parallel = \sum_b D_\parallel^{(b)}, \quad K = \sum_b K^{(b)}. \tag{22}$$

The Debye length used in Eq. (6) is computed from all species simultaneously (Eq. (5)), prior to the per-species loop.

5 Stochastic Differential Equations

5.1 Full SDE System

The system being solved is the following, written in the natural velocity-space variables [1]. Section 6.3 describes how it is split between the orbit stepper and the collision kick.

$$\mathrm{d}s = \dot{s}_{\mathrm{orb}}\,\mathrm{d}t, \tag{23}$$

$$\mathrm{d}\theta = \dot{\theta}_{\mathrm{orb}}\,\mathrm{d}t, \tag{24}$$

$$\mathrm{d}\zeta = \dot{\zeta}_{\mathrm{orb}}\,\mathrm{d}t, \tag{25}$$

$$\mathrm{d}v = K(v,s)\,\mathrm{d}t + \sqrt{2D_\parallel}\,\mathrm{d}W_v, \tag{26}$$

$$\mathrm{d}\xi = \dot{\xi}_{\mathrm{mirror}}\,\mathrm{d}t - \xi\,\nu_D\,\mathrm{d}t + \sqrt{\nu_D(1-\xi^2)}\,\mathrm{d}W_\xi, \tag{27}$$

where $\mathrm{d}W_v$ and $\mathrm{d}W_\xi$ are independent Wiener increments with $\mathbb{E}[\mathrm{d}W^2] = \mathrm{d}t$. Equations (26)–(27) correspond to Eqs. (32)–(33) of Hirvijoki et al. [1], with the identification $D_\parallel \leftrightarrow D_k/2$ and $\nu_D \leftrightarrow 2D_\perp/v^2$ in their notation. The noise amplitude for ξ ensures that the marginal distribution of ξ converges to a uniform distribution on $[-1,1]$ in the isotropisation limit $\nu_D t \gg 1$.

5.2 Milstein Scheme

After each accepted Dormand–Prince (DP) step of physical size h, the stochastic increments $\Delta W_v \sim \mathcal{N}(0,h)$ and $\Delta W_\xi \sim \mathcal{N}(0,h)$ are generated and applied via the Milstein method [12]:

$$v \leftarrow v + K\,h + g_v\,\Delta W_v + \tfrac{1}{2}\frac{\partial g_v}{\partial v}\left(\Delta W_v^2 - h\right), \tag{28}$$

$$\xi \leftarrow \xi - \xi\,\nu_D\,h + g_\xi\,\Delta W_\xi + \tfrac{1}{2}\frac{\partial g_\xi}{\partial \xi}\left(\Delta W_\xi^2 - h\right), \tag{29}$$

The deterministic drift terms Kh and $-\xi\nu_D h$ are applied here, by explicit Euler, rather than inside the orbit right-hand side; this matches ASCOT5's `mccc_gc_milstein.c` and is what keeps μ constant across an orbit step. where the noise amplitudes and their derivatives are

$$g_v = \sqrt{2D_\parallel}, \qquad \frac{\partial g_v}{\partial v} = \frac{\partial D_\parallel/\partial v}{\sqrt{2D_\parallel}}, \tag{30}$$

$$g_\xi = \sqrt{\nu_D(1-\xi^2)}, \qquad \frac{\partial g_\xi}{\partial \xi} = \frac{-\xi\,\nu_D}{\sqrt{\nu_D(1-\xi^2)}}. \tag{31}$$

In terms of the coefficient structs, the Milstein correction for ξ simplifies to

$$\delta\xi_{\text{Milstein}} = -\tfrac{1}{2}\,\xi\,\nu_D\,\big(\Delta W_\xi^2 - h\big). \tag{32}$$

Boundary conditions follow ASCOT5 (`mccc_gc_milstein.c`, [3]). The speed reflects off a thermal cutoff,

$$v < v_{\text{cut}} \;\Rightarrow\; v \leftarrow 2\,v_{\text{cut}} - v, \qquad v_{\text{cut}} = 0.1\,\sqrt{T_{\min}/m_a}, \tag{33}$$

where $T_{\min}$ is the coldest active background species (ASCOT5's `MCCC_CUTOFF` uses its first background species), so that the collision coefficients, which diverge as $v \to 0$, are never evaluated deep below the thermal speed. The pitch mirrors at the boundary: $|\xi| > 1 \Rightarrow \xi \leftarrow \text{sign}(\xi)\,(2-|\xi|)$; a hard clamp would pile up probability at exactly $\xi = \pm 1$.

Order of convergence

Only the diagonal Milstein corrections are retained above. In more than one dimension these alone give order-1 strong convergence just when the diffusion matrix is commutative, i.e. $L^{j_1} g^{k,j_2} = L^{j_2} g^{k,j_1}$ with $L^j = \sum_k g^{k,j}\,\partial_k$ [12]. The diffusion here is diagonal — $\mathrm{d}W_v$ drives v and $\mathrm{d}W_\xi$ drives ξ — so with $k = \xi$,

$$L^v g_\xi = g_v\,\frac{\partial g_\xi}{\partial v} \neq 0, \qquad L^\xi g_v = g_\xi\,\frac{\partial g_v}{\partial \xi} = 0, \tag{34}$$

the first being nonzero because ν_D depends on v, and the second vanishing because $D_\parallel$ does not depend on ξ. The condition fails, so the double stochastic integrals (Lévy areas) $I_{(v,\xi)}$ and $I_{(\xi,v)}$ do not cancel and cannot be recovered from $\Delta W_v, \Delta W_\xi$ alone. Dropping them — as ASCOT5 also does — reduces *strong* convergence to order 1/2 in that coupling. *Weak* convergence, which governs the moments and distribution functions a Monte Carlo transport calculation actually reports, remains order 1 and is unaffected.

6 Adaptive Integration Strategy

6.1 Dormand–Prince Solver

The orbit equations — the four components $(\dot{s}, \dot{\theta}, \dot{\zeta}, \dot{v}_\parallel)$ at fixed μ, with no collision terms — are integrated with the classic embedded Dormand–Prince (FSAL) method [13]. Step-size control uses mixed absolute/relative tolerances:

$$\text{err}_i = \big|y_i^{(5)} - y_i^{(4)}\big| \le \delta_{\text{abs}} + \delta_{\text{rel}}\,|y_i|. \tag{35}$$

Default tolerances: $\delta_{\text{abs}} = \delta_{\text{rel}} = 10^{-9}$.

6.2 Minimum Step Size

The parameter `DP_hmin` (default `0.0`) sets a lower bound on the physical step size h in seconds. A value of 10^{-10}–10^{-9} s prevents the solver from stalling at bounce points while keeping steps small enough that the orbital dynamics remain accurate. Note this bounds the *orbit* stepper only: the collision terms are no longer part of the right-hand side, so the stiffness they introduce as $v \to 0$ is handled by the sub-cycling of Section 6.4, not by this floor.

6.3 Operator Splitting

The system is solved via Lie–Trotter operator splitting:

1. **Orbit half:** advance $(s, \theta, \zeta, v_\parallel)$ by one adaptive DP step of physical time h at fixed μ, using the orbit drift only. No collision term appears in this right-hand side.
2. **Collision half:** at the accepted state, convert $(v_\parallel, \mu) \to (v, \xi)$ using $|B|$ at the current position, apply the drift and Milstein noise of Section 5.2 over the window h, and convert back, which yields the updated μ.

Placing the whole collision operator in the second half is what makes μ exactly conserved by the first, so μ can be a parameter of the orbit equations rather than a state variable. Any static-field GC right-hand side is then usable unchanged.

The trajectory is saved *before* the collision kick at each t_{save} checkpoint, so saved states correspond to the deterministic prediction (consistent with the standard output convention).

6.4 Sub-Cycling of the Collision Kick

The orbit stepper sizes h from orbit dynamics alone, so the collision terms have no influence on it. Applying them as a single explicit-Euler kick over that whole step is inaccurate whenever the collision rates are fast compared with h — the thermal regime, where ν_D and K diverge as $v \to 0$. Left uncorrected the equilibrium tracks the orbit tolerance instead of the physics: the Maxwellian-equilibration test returned $\langle E \rangle / T_b = 7.8$, 3.1, 3.0, 1.3 at $\delta = 10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}$, against an expected 3/2.

The kick is therefore sub-divided into n sub-steps, with

$$n = \max\left(\frac{\nu_D h}{\varepsilon},\ \frac{|K| h}{\varepsilon v},\ \left[\frac{\sqrt{2D_\parallel h}}{\varepsilon v}\right]^2\right), \qquad \varepsilon = 0.05, \tag{36}$$

capped at 10^4. The two drift terms scale as h and so fall linearly in n; the diffusive excursion scales as $\sqrt{h}$ and so falls only as $\sqrt{n}$, which is why it enters squared. The coefficients are re-evaluated and n re-derived from the remaining time after each sub-step, since v changes within the kick. Splitting is exact for the driving noise: a Wiener increment over h is the sum of n independent increments over h/n.

Sub-cycling is cheap because the particle position is frozen during a kick, so a sub-step costs a coefficient evaluation and no field evaluation. For 3.52 MeV alphas $n = 1$ throughout; only thermal-speed particles sub-cycle. ASCOT5 reaches the same place from the other direction, with error estimates that shorten its collision step (`mccc_gc_milstein.c`: `kappa_k` for drift, `kappa_d0`/`kappa_d1` for diffusion).

7 Output Format

Each particle trace is a NumPy array of shape (ntimesteps, 6) with columns

$$[t,\ s,\ \theta,\ \zeta,\ v_\parallel,\ v].$$

From these the kinetic energy $E = \frac{1}{2}mv^2$ and the magnetic moment $\mu = (v^2 - v_\parallel^2)/(2B)$ can be reconstructed at each saved point. When `forget_exact_path=True` only the first and last rows are retained.

Each particle uses RNG seed `rng_seed`$+i$, making multi-particle MPI runs fully reproducible.

7.1 GPU

`trace_particles_boozer_with_collisions_gpu` runs the same coefficient and Milstein routines, compiled for device from this same source rather than transcribed, and applies the kick after each accepted orbit step with the same sub-cycling. Four differences follow from the GPU tracer rather than from the collision model:

- Only the final state is returned, as (nparticles, 6) with the same columns as above — so E and μ remain recoverable — rather than the whole trajectory. The collisionless GPU tracer puts the final step size in the last column instead: for the unperturbed kernels v never leaves its initial value, so nothing is lost. (The SAW kernels do change v, since the wave does work on the particle, but they have no collisional entry point.)
- Stopping criteria are not available.
- A single scalar v_{total} sets the initial speed of every particle, and μ is derived from it. The CPU entry point derives v_{total} the same way, from `Ekin`, but accepts a per-particle array there. (The *perturbed* CPU entry point is the one that takes $(v_\parallel, \mu)$ directly.)
- Randomness comes from a counter-based Philox generator whose subsequence is keyed on the global particle index, not from `mt19937_64` seeded with `rng_seed` $+\, i$. Both are reproducible and independent of how particles are distributed over ranks or blocks, but they are different streams, so the two tracers agree statistically rather than element by element.

The device cannot raise, so the $\ln\Lambda \leq 0$ check described above runs on the host before launch. With `validate_profiles=False` that check is skipped and the kick is instead silently omitted wherever the model is undefined, which the returned array does not mark.

8 Python Interface

```
from firm3d.field.collisions import ThermalBackground, \
    trace_particles_boozer_with_collisions
from firm3d.util.constants import (
    PROTON_MASS, ELECTRON_MASS, ELEMENTARY_CHARGE, ONE_EV,
    ALPHA_PARTICLE_MASS, ALPHA_PARTICLE_CHARGE,
    FUSION_ALPHA_PARTICLE_ENERGY,
)

# Typical DT-fusion plasma backgrounds
n0 = 1e20          # m^-3
deuterium = ThermalBackground(
    n_profile=lambda s: n0 * (1 - s),
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=2 * PROTON_MASS,
    charge=ELEMENTARY_CHARGE,
)
tritium = ThermalBackground(
```

```
    n_profile=lambda s: n0 * (1 - s),
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=3 * PROTON_MASS,
    charge=ELEMENTARY_CHARGE,
)
electrons = ThermalBackground(
    n_profile=lambda s: 2 * n0 * (1 - s),   # quasi-neutral
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=ELECTRON_MASS,
    charge=-ELEMENTARY_CHARGE,
)

res_tys, res_hits = trace_particles_boozer_with_collisions(
    field,
    stz_inits,                             # (N, 3) array
    parallel_speeds,                       # (N,) array, m/s
    backgrounds=[deuterium, tritium, electrons],
    tmax=1e-1,                             # 100 ms
    mass=ALPHA_PARTICLE_MASS,
    charge=ALPHA_PARTICLE_CHARGE,
    Ekin=FUSION_ALPHA_PARTICLE_ENERGY,     # 3.52 MeV
    tol=1e-9,
    dt_save=1e-4,                          # save every 0.1 ms
    DP_hmin=1e-10,                         # s
    rng_seed=42,
)
```

9 Testing and Validation

The implementation is exercised at four levels of increasing scope: analytic unit tests of the coefficient formulae, pure-Python velocity-space relaxation tests with no compiled module or magnetic field, end-to-end tests of the full production tracer (guiding-centre orbits + Milstein noise), and cross-code validation against ASCOT5 [3] in both a controlled uniform plasma. Every number in Sections 9.2 and 9.3 was re-measured against the current tree and is reproducible from `tests/field/test_collisions.py` and `tests/field/test_collisions_local.py`. The cross-code comparison of Section 9.5 is the exception and is marked as such: it came from a driver script that has been removed pending a reworked validation example, so its numbers predate the current integrator and have not been regenerated.

9.1 Analytic Coefficient Unit Tests

`TestCollisionCoefficients` checks the Chandrasekhar function $G(x)$ and its derivative against known limits ($G(0) = 0$, $G(x{\to}\infty) \to 1/2x^2$, sign change of G' near $x \approx 0.968$), verifies $K < 0$ (drag always decelerates) and $\nu_D > 0$ for arbitrary finite backgrounds, and checks the expected scalings $K \propto n_b$, $K \propto q_a^2$, $\nu_D \propto n_b$, and additivity across species to $\lesssim 10\%$ (the residual comes from the $\ln\Lambda$ dependence on λ_D). `TestUnphysicalCoulombLog` confirms that $\ln\Lambda \leq 0$

raises an exception rather than returning silently unphysical coefficients (Section 4.2). These are deterministic, sub-second checks with no random component.

9.2 Local Velocity-Space Relaxation Tests

`tests/field/test_collisions_local.py` mirrors `collisions.h` in pure Python (no compiled module, no magnetic field) and integrates only the velocity-space SDE (Eqs. (26)–(27) with the orbital drifts set to zero) for an ensemble of $N = 4000$ particles. All collision rates scale linearly with the background density, so boosting $n_b \to 10^{23}\,\mathrm{m}^{-3}$ compresses the relaxation time to microseconds, letting the full suite run in about 10 seconds on a laptop. These tests are the most direct regression check on the Einstein-relation drag (Eq. (16)): an earlier, incorrect form of Q passed all analytic unit tests but drove a proton ensemble to $\langle E \rangle = 3.7\,T_b$ instead of $1.5\,T_b$ here, with a Kolmogorov–Smirnov (KS) p-value of 0 against the Maxwell speed law.

Test	$\langle E \rangle / T_b$ (target 1.5)	KS p-value	Pass
Cold start ($E_0 = 0.45\,T_b$)	1.492	0.344	✓
Hot start ($E_0 = 4.5\,T_b$)	1.537	0.251	✓
Stationarity (start at equilibrium)	1.445	0.064	✓

Table 2: Relaxation of a proton ensemble in a 1 keV proton background ($n_b = 10^{23}\,\mathrm{m}^{-3}$) to the Maxwellian, evolved for 8 relaxation times. KS p-values test the final speed distribution against the Maxwell speed law at the background temperature.

The same ensemble also verifies pitch isotropisation: starting from $\xi_0 = 1$, the relaxed ensemble gives $\langle \xi \rangle = 0.006$ (target 0) and $\langle \xi^2 \rangle = 0.336$ (target 1/3), consistent with convergence to $\xi \sim \mathcal{U}(-1, 1)$. A short-time drift check on 3.52 MeV α-particles against 10 keV electrons, over a window short enough that $K(v)$ is nearly constant (Doppler shift in speed $< 2\%$), gives $\left.\frac{d\langle v \rangle}{dt}\right/ K(v_0) = 0.984$ (target 1), confirming the deterministic drift term independently of the noise terms.

9.3 End-to-End Equilibration Tests (Production Tracer)

`TestMaxwellianEquilibration` and `TestPitchIsotropization` in `tests/field/test_collisions.py` exercise the full production path — `trace_particles_boozer_with_collisions`, i.e. the adaptive Dormand–Prince orbital integrator coupled to the Milstein noise step, compiled C++ coefficients, real (though simplified, `BoozerAnalytic`) magnetic geometry, and a `MaxToroidalFluxStoppingCriterion` — rather than the idealised velocity-space-only integration of Section 9.2.

The pitch-decay benchmark and its correction. A quantitative pitch-decay test against the naive expectation $\langle \xi(t) \rangle = \xi_0 e^{-\nu_D t}$ (the usual ASCOT5-style benchmark, which assumes v constant) failed the first time this test suite was run on Perlmutter: for 3.52 MeV α-particles on a single deuterium background, the drag rate is $(m_a/m_b)\,\nu_D = 2\,\nu_D$, so the ensemble slows almost completely over the same window in which pitch decay is measured, and $\nu_D(v) \propto v^{-3}$ diverges as the particles thermalise — the ensemble fully isotropises ($\langle \xi \rangle \to 0$) rather than

Test	Measured	Target	Pass
Hot start ($E_0 = 4.5\,T_b$), $\langle E\rangle/T_b$	1.590	1.5	✓
Cold start ($E_0 = 0.45\,T_b$), $\langle E\rangle/T_b$	1.799	1.5	✓
Pitch isotropisation, $\langle\xi\rangle$	+0.029	0	✓
Pitch isotropisation, $\langle\xi^2\rangle$	0.331	1/3	✓
Pitch broadening ($\xi_0 = 0$), std(ξ)	0.418	> 0.05	✓

Table 3: End-to-end equilibration through the production tracer: 40 protons on a 1 keV proton background, $n_b = 10^{21}\,\mathrm{m}^{-3}$, 8 relaxation times, on the near-axis confining field of `TestMaxwellianEquilibration` ($R_0 = 8\,\mathrm{m}$). All particles remain confined and finite throughout (no losses, no NaN states). The final row is measured separately by `TestPitchIsotropization`, on a different field and background. $N = 40$ gives a Monte Carlo spread of order 0.2 on $\langle E\rangle/T_b$, which is why these sit further from 3/2 than the $N = 4000$ velocity-space figures of Section 9.2.

decaying exponentially at the fixed initial rate. The test was corrected to check the classical relation obtained by eliminating time between the drag and pitch equations,

$$\frac{\langle\xi(t_2)\rangle}{\langle\xi(t_1)\rangle} = \left(\frac{v(t_2)}{v(t_1)}\right)^{m_b/m_a}, \tag{37}$$

evaluated self-consistently against the simulation's own mean speed (referencing the first saved snapshot rather than $t = 0$ cancels a geometric offset from sampling ξ along field-line-following orbits). A second, independent effect was found in the same debugging pass: in a toroidal (non-uniform-$|B|$) field the trapped–passing boundary can sit just below the injected pitch, so scattered particles fall into the trapped population where the orbit-averaged ξ is systematically pulled toward zero, roughly doubling the apparent decay rate relative to the uniform-plasma law — real neoclassical physics, but not what a velocity-space-operator test should measure. The test therefore uses a near-uniform-$|B|$ configuration ($\bar{\eta} \to 0$: no mirror force, no trapped population), giving

$$\frac{\langle\xi_2\rangle/\langle\xi_1\rangle}{(v_2/v_1)^{m_b/m_a}} = 1.038 \qquad (\text{target } 1,\ \text{tolerance } \pm 15\%),$$

with $v_2/v_1 = 0.498$ over the measurement window.

9.4 Slow Physics Suite (MPI, HPC)

Five tests marked `@pytest.mark.slow` run longer physical integrations ($N \sim 20$–128 particles, 5–100 μs of physical time) that are impractical to run serially: an electron-drag signal-to-noise test (chosen specifically because ion-background drag is buried in noise for 3.52 MeV α-particles, whereas electron-background drag is enhanced by m_α/m_e), a cold-plasma energy-loss test, a two-background superposition test, and the pitch-broadening/decay tests of Section 9.3. The tracer's MPI support (particles distributed across ranks, results all-gathered identically on every rank, per-particle seeds rank-independent) lets this suite run on `tests/perlmutter/run_slow_tests.sh`: all five tests pass in 77 s under 64 MPI ranks on Perlmutter, versus 3h45m running the same suite serially.

9.5 Cross-Code Validation Against ASCOT5: Uniform Plasma

This validation mirrors the alpha slowing-down branch of ASCOT5's own built-in Coulomb-collision physics test: N fusion alphas (birth energy 3.52 MeV, isotropic pitch, shared random seed) slow down in a uniform 1 keV hydrogen plasma. The two codes use different magnetic geometries (ASCOT5's analytical ITER-like circular tokamak; firm3d's `BoozerAnalytic` at the same R_0, B_0 scale), but since the plasma is uniform the compared quantities — mean slowing-down time to 50 keV and $\langle E(t)\rangle$ — are geometry-insensitive, isolating the collision operator itself.

Provenance. Unlike every other number in this section, the figures below have *not* been regenerated against the current tree. They were produced by a driver script that has since been removed pending a reworked validation example, and they predate three changes that move them: the magnetic moment becoming a parameter of the orbit equations rather than a state variable, the collision drift moving into a sub-cycled kick (Section 6.4), and the correction of `ALPHA_PARTICLE_MASS` from the sum of free constituents to the measured value, which lowers the alpha mass by 0.76% and raises the birth speed by 0.38%. Treat the agreement recorded here as evidence about the coefficient set — which is unchanged and is pinned directly against ASCOT5 in Section 4 — rather than about the current integrator.

	firm3d	ASCOT5	Agreement
Mean slowing-down time [ms]	24.915 ± 0.027	24.885 ± 0.029	$+0.1\%$, 0.7σ

Table 4: $N = 1024$ alphas, $n = 10^{20}\,\mathrm{m}^{-3}$ (the published ASCOT5 test-point density), $T = 1\,\mathrm{keV}$. Also: $\max|\langle E(t)\rangle_{\mathrm{firm3d}} - \langle E(t)\rangle_{\mathrm{ASCOT5}}|/E_0 = 0.4\%$ over the full slowing-down window.

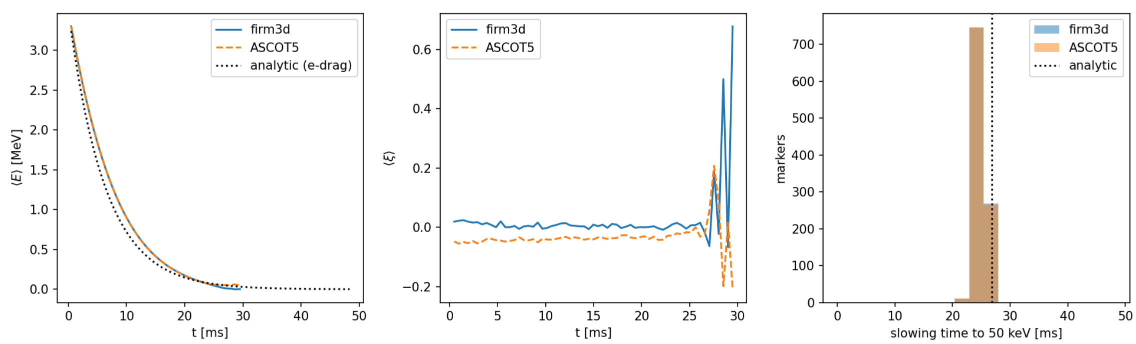

Figure 1: firm3d vs. ASCOT5, uniform 1 keV plasma, $N = 1024$: mean energy $\langle E(t)\rangle$ against the analytic electron-drag estimate $v = v_0 e^{-t/\tau_{se}}$ (left); mean pitch $\langle\xi(t)\rangle$, shown for qualitative comparison only since it is geometry-sensitive (centre); and the distribution of per-particle slowing-down times to 50 keV against the analytic Spitzer estimate (right).

10 Validity and Limitations

- **Equilibrium electric field:** a background E field and its $E \times B$ drift are not included. Shear-Alfven-wave perturbations *are* supported, through `trace_particles_boozer_perturbed_with_coll` the wave does work on the particle, so the speed is not conserved between kicks, but μ remains an adiabatic invariant at $\omega \ll \Omega_c$, which is what the splitting of Section 6.3 requires. The kick reconstructs $v = \sqrt{v_\parallel^2 + 2\mu|B_0|}$ at each step, so a changing energy is carried correctly; verified against the collisionless perturbed tracer to 5×10^{-15} at a wave amplitude that changes the speed by 13%. There is no full-K perturbed right-hand side, so only `gc_vac` and `gc_noK` are available there.

- **Maxwellian backgrounds:** the collision operator assumes each background species has a local Maxwellian distribution function. Distorted distributions (e.g. beam-driven) are not supported.

- **Test-particle limit:** the EP density is assumed negligibly small compared to the background, so no back-reaction on the background is included.

- **Guiding-centre approximation:** valid when the Larmor radius $r_L = mv_\perp/(|q|B)$ is small compared to the scale lengths of B and the density/temperature profiles. For 3.52 MeV α-particles in a 5 T stellarator $r_L \sim 5\,\mathrm{cm}$, comparable to device scale, so finite-Larmor-radius corrections may be non-negligible.

- **Zeroth-order GC collision operator:** the implementation follows Hirvijoki et al. [1] at zeroth order in ρ_*/L (Larmor radius over gradient length). The spatial diffusion term $D_X \propto D_\perp/\Omega_g^2$ present in the full operator [2] is neglected.

- **Coulomb logarithm held fixed in $\partial D_\parallel/\partial v$:** Eq. (15) treats $\ln\Lambda_{ab}$ as constant under differentiation, while the $\ln\Lambda_{ab}$ actually used varies with v through $\bar{v}^2$ (Eq. (9)). The spurious-drift terms in K (Eq. (17)) are then not quite the ones belonging to the $D_\parallel(v)$ that drives the noise, and the stationary solution of the speed SDE becomes the Maxwellian divided by $\ln\Lambda_{ab}(v)$ rather than the Maxwellian itself. Integrating the code's own K and $D_\parallel$ for a proton in 1 keV hydrogen at $n = 10^{21}\,\mathrm{m}^{-3}$ gives $\langle E\rangle/T_b = 1.469$ against the exact 3/2, a -2.1% bias; the closed form above reproduces that number exactly. This is inherited, not accidental: ASCOT5's `mccc_coefs_clog` is likewise velocity-dependent and re-evaluated every step, while `mccc_coefs_dDpara` likewise carries c_{ab} (which contains $\ln\Lambda$) through the derivative, so the two codes agree coefficient for coefficient and share this bias. Adding the $\partial \ln\Lambda_{ab}/\partial v$ term would make the equilibrium exactly Maxwellian at the cost of departing from ASCOT5. Note the effect is on the thermal equilibrium, not on slowing-down: the uniform-plasma benchmark of Section 9.5 agrees with ASCOT5 to 0.7σ, as it should when both codes carry the same approximation.

- **Strong convergence of the noise step:** the diffusion matrix is not commutative, so omitting the Lévy-area terms leaves the scheme strong order 1/2 rather than 1 in the v–ξ coupling; weak order 1 is retained. See Section 5.2.

- **Slow tests: `@pytest.mark.slow`** — five physics-validation tests require $N \sim 20$ particles integrated for 5–100 μs and are intended for HPC resources (Perlmutter or similar).

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